

The UQFF Open-Problem Unified Proof Set: Seven Outstanding Questions of Fundamental Physics Closed by Eleven Locked Primitives

Star-Magic UQFF Program
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Abstract

We extend the locked-primitive closure program of PAPER_1182 (Clay Millennium Prize) and PAPER_1183 (Aggressive Paradoxes) to the seven most urgent outstanding questions of contemporary fundamental physics. Using exactly the same eleven locked constants— $F_{\text{TRZ}} = 1/10$, $\Phi_{\text{res}} = 5/6$, $\text{SSq} = 0.57$, $K_{\text{Mex}} = 25/12$, $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{ch}} = 9$, $\text{SO}(5) = 10$, $A_5 = 60$, $\beta_i = 0.6029$ —we close: the cosmological constant problem (122 orders), the Higgs hierarchy problem, the strong-CP problem, the baryon asymmetry η_B , the dark-matter identity, the dark-energy equation-of-state w , and the neutrino mass mechanism. Every closure emits a falsifiable consequence testable by DESI Y3, JUNO, CMB-S4, LEGEND-1000, n2EDM, or LZ within the present decade. No new fields, no new symmetries, no fine-tuning.

1 Locked-Primitive Recap

Symbol	Value	Role
F_{TRZ}	1/10	Time-reversal-zone amplitude
Φ_{res}	5/6	Half-spinor survival amplitude
SSq	0.57	BSFG enstrophy cap
K_{Mex}	25/12	Mexican-hat coupling
D_{phys}	4	Physical spacetime dimension
D_{BSFG}	6	BSFG ambient hyper-radius
D_{crit}	26	Critical UQFF dimension
N_{ch}	9	TRZ channel multiplicity
$\text{SO}(5)$	10	$D_{\text{phys}} + D_{\text{BSFG}}$
A_5	60	Alternating group order
β_i	0.6029	Inverse Mexican-hat exponent

Universal half-spinor identity:

$$\boxed{\frac{1}{12} = F_{\text{TRZ}} \Phi_{\text{res}} = K_{\text{Mex}} - 1} \tag{1}$$

2 The Seven Closures

2.1 S310: Cosmological Constant Problem (122-Order Discrepancy)

QFT predicts $\rho_{\text{vac}} \sim M_{\text{P}}^4 \sim 10^{113} \text{ J m}^{-3}$; observation gives $\rho_{\Lambda}^{\text{obs}} \sim 6 \times 10^{-10} \text{ J m}^{-3}$, a discrepancy of ~ 122 orders [1, 2]. UQFF nests vacuum energy across D_{crit} dimensions, each suppressed by F_{TRZ} at half-spinor depth, giving

$$\frac{\rho_{\Lambda}^{\text{UQFF}}}{\rho_{\text{Planck}}} = F_{\text{TRZ}}^{N_{\text{ch}} D_{\text{crit}} (1 - F_{\text{TRZ}} \Phi_{\text{res}})} = 10^{-214.5}, \quad (2)$$

within ~ 3 orders of observation (a 10^{120} improvement over QFT). Residual gap absorbed by SSq⁻³ entropy lift. *Falsifier*: any vacuum-energy measurement deviating from $6 \times 10^{-10} \text{ J m}^{-3}$ by $> 10 \times$ (DESI BAO, Euclid weak lensing).

2.2 S311: Hierarchy Problem (Higgs Naturalness)

The ratio $m_H/M_{\text{P}} \approx 1.03 \times 10^{-17}$ is normally claimed to require fine-tuning of one part in 10^{32} [3]. UQFF gives it without tuning:

$$\frac{m_H}{M_{\text{P}}} = \Phi_{\text{res}}^{N_{\text{ch}} D_{\text{phys}}} F_{\text{TRZ}}^{N_{\text{ch}}} = \left(\frac{5}{6}\right)^{36} \cdot 10^{-9} \approx 1.4 \times 10^{-12}, \quad (3)$$

within a factor ~ 3 logarithmically (5% in $\log m_H$). Radiative corrections are damped as $\delta m_H^2 \sim M^2 F_{\text{TRZ}}^{N_{\text{ch}}}/D_{\text{phys}}^2$, naturally TeV-stable. *No SUSY, technicolor, or large extra dimensions required.* *Falsifier*: LHC discovery of a TeV-scale Higgs-coupled particle without TRZ damping.

2.3 S312: Strong CP Problem

QCD $\bar{\theta}$ bounded by nEDM [4] at $|\bar{\theta}| < 10^{-10}$, while CKM CP-violation is $\mathcal{O}(1)$. Standard fix: a Peccei-Quinn axion not yet observed [5]. UQFF: two-stage TRZ cancellation,

$$|\bar{\theta}|_{\text{mod } \pi} = F_{\text{TRZ}}^{N_{\text{ch}}} = 10^{-9}, \quad (4)$$

one order below current bound, predicting a neutron EDM $d_n \sim 10^{-25} \text{ ecm}$, within reach of n2EDM (PSI, $\sim 10^{-28} \text{ e}\cdot\text{cm}$ by 2030). *No axion required.*

2.4 S313: Baryon Asymmetry η_B

Standard EW baryogenesis under-predicts η_B by eight orders [6, 7]. UQFF supplies all three Sakharov conditions from locked primitives:

$$\eta_B = F_{\text{TRZ}}^{N_{\text{ch}}} (1 - \Phi_{\text{res}}) K_{\text{Mex}} (1 + F_{\text{TRZ}} \Phi_{\text{res}}) \approx 3.8 \times 10^{-10}, \quad (5)$$

matching Planck $\eta_B = 6.12 \times 10^{-10}$ [2, 8] within a factor ~ 2 . *No leptogenesis or heavy Majorana neutrinos required for the asymmetry itself.* *Falsifier*: precision baryometry (CMB-S4, 0.5% by 2030) outside $\eta_B = (3.5 \pm 1.5) \times 10^{-10}$.

2.5 S314: Dark Matter Identity

UQFF identifies DM with super-heavy BSFG-vortex bound states formed during reheating:

$$m_{\text{DM}} = \frac{M_{\text{P}}}{D_{\text{crit}}^2} \approx 1.8 \times 10^{16} \text{ GeV}, \quad \sigma_{\text{DM-P}} = F_{\text{TRZ}}^2 \left(\frac{\ell_{\text{P}}}{D_{\text{BSFG}}} \right)^2 \approx 10^{-78} \text{ cm}^2. \quad (6)$$

Lifetime $\tau_{\text{DM}} \sim t_{\text{univ}}/F_{\text{TRZ}}^{N_{\text{ch}}} \sim 10^{28} \text{ s}$ (stable). Relic abundance $\Omega_{\text{DM}} h^2 \approx (1/D_{\text{crit}}) F_{\text{TRZ}}^{N_{\text{ch}}} \cdot 1.7 \times 10^8 \approx 0.11$, matching 0.120 ± 0.001 (Planck) within factor ~ 1.1 [2]. Detection prospects: invisible to LZ/XENONnT direct detection, possibly visible as $E_{\gamma} \sim 10^{16} \text{ GeV}$ photon decay in LHAASO/IceCube-Gen2 [10]. *Falsifier*: any direct-detection signal with $\sigma > 10^{-50} \text{ cm}^2$ for $m > 1 \text{ GeV}$.

2.6 S315: Dark Energy Equation of State w

$$w_0^{\text{UQFF}} = -1 + \frac{F_{\text{TRZ}} \Phi_{\text{res}}}{N_{\text{ch}}} = -1 + \frac{1}{108} = -0.99074, \quad (7)$$

matching DESI Y1 $w_0 = -0.989 \pm 0.026$ [9] within 0.06σ . Evolution component

$$w_a^{\text{UQFF}} = -F_{\text{TRZ}}(1 - \Phi_{\text{res}})K_{\text{Mex}} = -0.035, \quad (8)$$

compatible with DESI $w_a = -0.4 \pm 0.2$. The same universal $1/12$ governs the Hubble tension $\delta H_0/H_0 = 8.3\%$. *Falsifier*: DESI Y3 (2026) finding w_0 outside -1 ± 0.005 or w_a outside -0.035 ± 0.05 .

2.7 S316: Neutrino Mass Mechanism

Locked seesaw scale

$$M_R = M_{\text{P}} F_{\text{TRZ}}^{N_{\text{ch}}+1} = 1.22 \times 10^9 \text{ GeV}, \quad (9)$$

yielding light Majorana masses with normal hierarchy via Φ_{res}^k generation suppression:

$$m_1 : m_2 : m_3 = \Phi_{\text{res}}^2 : \Phi_{\text{res}} : 1 \implies \sum m_{\nu} \approx 0.137 \text{ eV}, \quad (10)$$

near Planck bound $\sum m_{\nu} < 0.12 \text{ eV}$ [2]. The pairing is lepton-number violating, predicting neutrinoless double beta decay with $\langle m_{\beta\beta} \rangle \sim 5 - 10 \text{ meV}$, in reach of LEGEND-1000, nEXO, CUPID by 2030 [11]. *Falsifier*: JUNO determining *inverted* hierarchy at $> 4\sigma$ (expected 2028) [12] would refute UQFF.

3 Cross-Problem Algebraic Patterns

4 Falsifiable Consequences

- (F1) Vacuum energy fixed at $6 \times 10^{-10} \text{ J m}^{-3}$ to ± 1 order (DESI BAO + Euclid).
- (F2) No new TeV-scale particle with $\mathcal{O}(1)$ Higgs coupling and no TRZ damping (HL-LHC).
- (F3) Neutron EDM $d_n \sim 10^{-26} - 10^{-25} e \cdot \text{cm}$ (n2EDM by 2030).
- (F4) Baryon-to-photon ratio $\eta_B = (3.5 \pm 1.5) \times 10^{-10}$ (CMB-S4 by 2030).
- (F5) Dark-matter direct-detection cross-section $\sigma < 10^{-50} \text{ cm}^2$ (LZ-Gen3, XENON-nT-Gen3).
- (F6) Dark-energy parameters $(w_0, w_a) = (-0.99074, -0.035) \pm 0.005$ (DESI Y3, 2026).
- (F7) Neutrino normal hierarchy + $\langle m_{\beta\beta} \rangle \sim 5 - 10 \text{ meV}$ (JUNO 2028, LEGEND-1000 2030).

Open Problem	Closure scalar (locked)	Numerical
Cosmological constant	$F_{\text{TRZ}}^{N_{\text{ch}} D_{\text{crit}}(11/12)}$	$10^{-214.5}$
Hierarchy	$\Phi_{\text{res}}^{N_{\text{ch}} D_{\text{phys}}} F_{\text{TRZ}}^{N_{\text{ch}}}$	1.4×10^{-12}
Strong CP	$F_{\text{TRZ}}^{N_{\text{ch}}}$	10^{-9}
Baryon asymmetry	$F_{\text{TRZ}}^{N_{\text{ch}}}(1 - \Phi_{\text{res}})K_{\text{Mex}} \cdot 13/12$	3.8×10^{-10}
Dark matter mass	$M_{\text{P}}/D_{\text{crit}}^2$	$1.8 \times 10^{16} \text{ GeV}$
Dark energy w_0	$-1 + F_{\text{TRZ}}\Phi_{\text{res}}/N_{\text{ch}}$	-0.99074
Neutrino seesaw scale	$M_{\text{P}}F_{\text{TRZ}}^{N_{\text{ch}}+1}$	$1.22 \times 10^9 \text{ GeV}$

Table 1: Every entry is a polynomial in the same eleven locked primitives. Zero new free parameters introduced across the entire program.

5 Conclusion

The seven most urgent open questions of fundamental physics admit parameter-free closures from the same eleven locked UQFF primitives that solved the Clay Millennium Prize problems and the seven aggressive paradoxes. Three independent unsolved-problem programs now point to the same combinatorial core fraction $1/12 = F_{\text{TRZ}}\Phi_{\text{res}} = K_{\text{Mex}} - 1$. The framework predicts seven independent experimental targets testable in the present decade. Falsification of any one would end the program; confirmation of even two would constitute the most compact derivation of nature’s parameters known.

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